TA Session 8: Statistics for ML

Point Estimation, Sample Mean and Sample Variance

Sheldon M. Ross, *Introduction to Probability and Statistics for Engineers and Scientists*, Fifth Edition (2014)

Part A — Sample Mean and Point Estimation

Q1. Normal probability and the standard-normal table

Ch. 5, Example 5.5a, pp. 172–173; medium

If X is normally distributed with mean $\mu = 3$ and variance $\sigma^2 = 16$, use the standard-normal probability table to find

(a) $P\{X < 11\}$;

(b) $P\{X > -1\}$;

(c) $P\{2 < X < 7\}$.

For each part, express the result in terms of $Z = (X - \mu)/\sigma$, and sketch or mark the corresponding probability region on a normal curve.

Q2. Mean of two equal-sized groups

Ch. 2, Problem 11, p. 45

A set of 198 observations has mean 120 for its first 99 observations and mean 100 for its final 99 observations.

(a) Find the sample mean of all 198 observations.

(b) Can the overall sample median and mode be determined from the information given? Explain.

Q3. Estimating mean ash percentage

Ch. 2, Problem 13(a), p. 46

Twelve measurements of ash percentage in coal are

$$9.2,\ 14.1,\ 9.8,\ 12.4,\ 16.0,\ 12.6,\ 22.7,\ 18.9,\ 21.0,\ 14.5,\ 20.4,\ 16.9.$$

Find the sample mean ash percentage.

Q4. Can state-level means simply be averaged?

Ch. 2, Problem 15, p. 46

Suppose that the average pay of workers is known for each of the 50 US states.

(a) Must the ordinary average of the 50 state averages equal the mean pay for all workers in the United States?

(b) What additional information is required to calculate the overall mean pay? Give the correct expression.

Q5. Mean and variance of sample averages

Ch. 6, Problem 1(b),(d), p. 227

Let X_1, X_2, X_3 be independent random variables, each with

$$P\{X_i = 0\} = 0.2, \quad P\{X_i = 1\} = 0.3, \quad P\{X_i = 3\} = 0.5.$$

Put $\bar{X}_2 = (X_1 + X_2)/2$ and $\bar{X}_3 = (X_1 + X_2 + X_3)/3$. Determine $E[\bar{X}_2]$, $\mathrm{Var}(\bar{X}_2)$, $E[\bar{X}_3]$, and $\mathrm{Var}(\bar{X}_3)$.

Part B — Sample Variance (Known and Unknown Mean)

Q6. Equal means, unequal variances

Ch. 2, Example 2.3f, pp. 22–23

For $A = (3, 4, 6, 7, 10)$ and $B = (-20, 5, 15, 24)$, calculate the sample mean and sample variance of each. Which set shows more variation?

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Q7. Recovering two observations

Ch. 2, Problem 14, p. 46

Five observations have sample mean $\bar{x} = 104$ and sample variance $s^2 = 16$. Three values are 102, 100, and 105. Find the other two observations.

Q8. Sample mean and sample variance from university data

Ch. 2, Problem 21(a),(b), p. 48

The following data record how many months 12 universities appeared among the ten most searched universities:

$$114, 114, 114, 113, 111, 97, 94, 66, 63, 52, 48, 33.$$

Find (a) the sample mean and (b) the sample variance.

Q9. Sample Mean and Variance from Grouped Data

Ch. 2, Problem 17(b), (c), (e), pp. 46–47

An experiment measuring the percentage shrinkage on drying of 50 clay specimens produced the following observations:

18.2	21.2	23.1	18.5	15.6
20.8	19.4	15.4	21.2	13.4
16.4	18.7	18.2	19.6	14.3
16.6	24.0	17.6	17.8	20.2
17.4	23.6	17.5	20.3	16.6
19.3	18.5	19.3	21.2	13.9
20.5	19.0	17.6	22.3	18.4
21.2	20.4	21.4	20.3	20.1
19.6	20.6	14.8	19.7	20.5
18.0	20.8	15.8	23.1	17.0

(a) Calculate the sample mean of the observations.

(b) Calculate the sample variance using

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

(c) Group the observations into class intervals of width 1 percentage point, starting at 13.0. Assume all observations in each interval are located at its midpoint. Calculate the sample mean and sample variance of the grouped data.

(d) Compare the original and grouped results. Explain why the sample means and sample variances may differ.

Q10. Why is the denominator $n - 1$?

Ch. 6, §6.4, pp. 219–220 (textbook derivation)

For a random sample $X_1, \ldots, X_n$ from a population with mean μ and variance σ^2, Ross defines

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Following the textbook's derivation, show that $E[S^2] = \sigma^2$.

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Worked solutions and highlighted lecture concepts

Q1. Solution

Ross Ch. 5, Example 5.5a

Key Concept

The **standard normal variable** is $Z = (X-\mu)/\sigma \sim N(0,1)$. In Ross book, $\Phi(z) = P\{Z \le z\}$. A normal-table entry is the **area to the left** of z; symmetry gives $\Phi(-z) = 1 - \Phi(z)$. These are direct normal probabilities, *not* CLT approximations.

Here $\sigma = \sqrt{16} = 4$, so $Z = (X-3)/4$.

$$\text{(a)} \quad P\{X < 11\} = P\left\{Z < \frac{11-3}{4}\right\} = \Phi(2) = \boxed{0.9772},$$

$$\text{(b)} \quad P\{X > -1\} = P\left\{Z > \frac{-1-3}{4}\right\} = P\{Z > -1\} = \Phi(1) = \boxed{0.8413},$$

$$\begin{aligned}\text{(c)} \quad P\{2 < X < 7\} &= P\{-0.25 < Z < 1\} \\ &= \Phi(1) - \Phi(-0.25) \\ &= 0.8413 - [1 - 0.5987] = \boxed{0.4400}.\end{aligned}$$

Shading guide: (a) shade to the left of $z = 2$; (b) shade to the right of $z = -1$; (c) shade between $z = -0.25$ and $z = 1$.

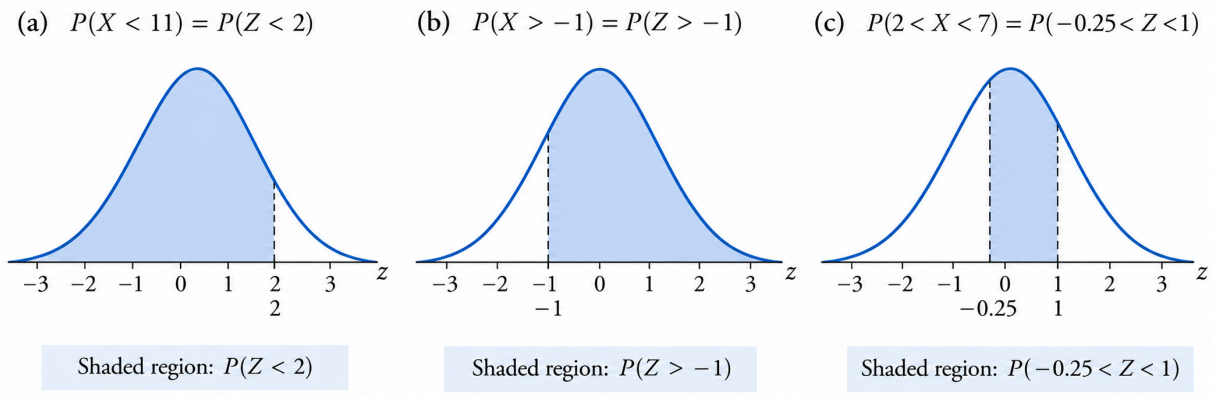


Figure 1: Standard normal curves showing the shaded probability regions for Q1(a), Q1(b), and Q1(c).

Q2. Solution

Ross Ch. 2, Problem 11

Key Concept

The mean of combined samples is a **weighted mean**. Equal sample sizes give equal weights. The mean does not, in general, determine the median or mode.

$$\bar{x} = \frac{99(120) + 99(100)}{198} = \boxed{110}.$$

The combined median cannot be uniquely determined (Ross notes that it lies between 100 and 120), and neither can the mode. The same subgroup mean can occur for many different sets of observations.

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Q3. Solution

Ross Ch. 2, Problem 13(a)

Key Concept

The observed **sample mean** $\bar{x}$ estimates the population parameter μ. The estimator $\bar{X}$ is random; its realized value $\bar{x}$ is a number.

$$\sum_{i=1}^{12} x_i = 188.5, \qquad \boxed{\bar{x} = \frac{188.5}{12} \approx 15.7083\%}.$$

Q4. Solution

Ross Ch. 2, Problem 15

Key Concept

When combining group means, use **sample-size weights**. An unweighted average of means is generally not the pooled mean.

No. Let n_j denote the number of workers in state j and $\bar{x}_j$ its mean pay. Then

$$\boxed{\bar{x}_{\text{all}} = \frac{\sum_{j=1}^{50} n_j \bar{x}_j}{\sum_{j=1}^{50} n_j}}.$$

We require the workforce sizes n_j (or equivalent proportions), not just the 50 means.

Q5. Solution

Ross Ch. 6, Problem 1(b),(d)

Key Concept

For i.i.d. observations, $E[\bar{X}_n] = \mu$ and $\text{Var}(\bar{X}_n) = \sigma^2/n$. **Independence** is needed when adding variances. A point estimator is **unbiased** if $E[\hat{\theta}] = \theta$. Its **bias** is $E[\hat{\theta}] - \theta$ and $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$.

$$\begin{aligned} \mu = E[X_i] &= 0(.2) + 1(.3) + 3(.5) = 1.8, \\ E[X_i^2] &= 0 + 1(.3) + 9(.5) = 4.8, \\ \sigma^2 = \text{Var}(X_i) &= 4.8 - (1.8)^2 = 1.56. \end{aligned}$$

Consequently,

$$\boxed{E[\bar{X}_2] = E[\bar{X}_3] = 1.8}, \qquad \boxed{\text{Var}(\bar{X}_2) = 0.78}, \qquad \boxed{\text{Var}(\bar{X}_3) = 0.52}.$$

Note: Both averages are unbiased for μ, so their MSEs are 0.78 and 0.52 respectively. This last observation follows from the lecture's bias/MSE definitions; the numerical task itself is Ross Problem 1(b),(d).

Q6. Solution

Ross Ch. 2, Example 2.3f

Key Concept

In Ross book, defines the **sample variance** as $s^2 = \sum_i (x_i - \bar{x})^2/(n-1)$. Both data sets can share a mean while having very different dispersion.

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$$\bar{x}_A = \frac{3+4+6+7+10}{5} = 6, \qquad s_A^2 = \frac{9+4+0+1+16}{4} = \boxed{7.5},$$
$$\bar{x}_B = \frac{-20+5+15+24}{4} = 6, \qquad s_B^2 = \frac{676+1+81+324}{3} = \boxed{360.67 \text{ (approx.)}}.$$

Data set B has greater sample variability.

Q7. Solution

Ross Ch. 2, Problem 14

Key Concept

With the **unknown population mean**, deviations are measured from $\bar{x}$, and the sample variance uses denominator $n-1$. Also, $\sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - n\bar{x}^2$ (See notes).

Let the missing values be u, v. Since $5\bar{x} = 520$,

$$u + v = 520 - (102 + 100 + 105) = 213.$$

Since $(n-1)s^2 = 64$,

$$(102-104)^2 + (100-104)^2 + (105-104)^2 + (u-104)^2 + (v-104)^2 = 64,$$

so $(u-104)^2 + (v-104)^2 = 43$. With $a = u - 104$, $b = v - 104$, we have $a + b = 5$ and $a^2 + b^2 = 43$, hence $ab = -9$. Therefore

$$a, b = \frac{5 \pm \sqrt{61}}{2}, \qquad \boxed{\{u, v\} = \left\{\frac{213+\sqrt{61}}{2}, \frac{213-\sqrt{61}}{2}\right\}}.$$

Approximately, the missing observations are 110.405 and 102.595.

Q8. Solution

Ross Ch. 2, Problem 21(a),(b)

Key Concept

For observations x_i, $\bar{x} = (\sum_i x_i)/n$ and $s^2 = (\sum_i x_i^2 - n\bar{x}^2)/(n-1)$. The denominator $n-1$ corrects the bias due to estimating μ using $\bar{X}$.

Here $n = 12$, $\sum_i x_i = 1019$, and $\sum_i x_i^2 = 96745$.

$$\boxed{\bar{x} = \frac{1019}{12} = 84.9167},$$

$$s^2 = \frac{96745 - (1019)^2/12}{11} = \boxed{928.6288 \text{ (approx.)}}.$$

Q9. Solution

Ross Ch. 2, Problem 17(b), (c), (e)

Key Concept

The **sample mean** and **sample variance** summarize the central tendency and dispersion of observations. For grouped data, class midpoints are used to approximate individual observations.

(a) Sample mean of the original data. For $n = 50$ observations, the sum is

$$\sum_{i=1}^{50} x_i = 948.9.$$

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Therefore,

$$\bar{x} = \frac{1}{50}\sum_{i=1}^{50} x_i = \frac{948.9}{50} = \boxed{18.978}.$$

(b) Sample variance of the original data. By definition,

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Expanding the squared deviations,

$$\begin{aligned}\sum_{i=1}^{n}(x_i - \bar{x})^2 &= \sum_{i=1}^{n} x_i^2 - 2\bar{x}\sum_{i=1}^{n} x_i + n\bar{x}^2 \\ &= \sum_{i=1}^{n} x_i^2 - n\bar{x}^2,\end{aligned}$$

where $\sum_{i=1}^{n} x_i = n\bar{x}$. For the data,

$$\sum_{i=1}^{50} x_i^2 = 18314.61.$$

It follows that

$$\begin{aligned}s^2 &= \frac{18314.61 - 50(18.978)^2}{49} \\ &= \frac{18314.61 - 18008.2242}{49} \\ &= \frac{306.3858}{49} \\ &= \boxed{6.2528 \text{ (approximately)}}.\end{aligned}$$

(c) Grouped-data estimates. Use intervals $[13,14), [14,15), \ldots, [24,25)$, with f_j representing the class frequency and m_j the midpoint.

Class interval	m_j	f_j	$f_j m_j$	$f_j m_j^2$
$[13,14)$	13.5	2	27.0	364.50
$[14,15)$	14.5	2	29.0	420.50
$[15,16)$	15.5	3	46.5	720.75
$[16,17)$	16.5	3	49.5	816.75
$[17,18)$	17.5	6	105.0	1837.50
$[18,19)$	18.5	7	129.5	2395.75
$[19,20)$	19.5	7	136.5	2661.75
$[20,21)$	20.5	10	205.0	4202.50
$[21,22)$	21.5	5	107.5	2311.25
$[22,23)$	22.5	1	22.5	506.25
$[23,24)$	23.5	3	70.5	1656.75
$[24,25)$	24.5	1	24.5	600.25
Total		**50**	**953.0**	**18494.50**

Replacing each observation by the midpoint of its class gives

$$\bar{x}_{\text{grouped}} = \frac{\sum_j f_j m_j}{\sum_j f_j} = \frac{953.0}{50} = \boxed{19.06}.$$

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The sample variance calculated from these approximate observations is

$$\begin{aligned} s^2_{\text{grouped}} &= \frac{1}{50-1}\left(\sum_j f_j m_j^2 - 50\bar{x}^2_{\text{grouped}}\right) \\ &= \frac{18494.50 - 50(19.06)^2}{49} \\ &= \frac{18494.50 - 18164.18}{49} \\ &= \frac{330.32}{49} \\ &= \boxed{6.7412 \text{ (approximately)}}. \end{aligned}$$

(d) Comparison and explanation.

	Original data	Grouped data
$\bar{x}$	18.978	19.060
s^2	6.2528	6.7412

The grouped calculations replace each measured value by its class midpoint. This changes both $\sum x_i$ and $\sum x_i^2$ and, consequently, the calculated mean and variance. Class midpoints provide approximations, not exact substitutes for the individual observations. The grouped variance happens to be larger here, but grouping does not always increase the sample variance.

Key result

The sample variance uses the denominator $n-1$:

$$s^2 = \frac{\sum_{i=1}^n x_i^2 - n\bar{x}^2}{n-1}.$$

For grouped data, x_i is replaced by the relevant class midpoint, with each midpoint repeated according to its frequency.

Q10. Solution

Ross Ch. 6, §6.4

Key Concept

The sample mean estimates unknown μ: $E[\bar{X}] = \mu$, $\text{Var}(\bar{X}) = \sigma^2/n$. When the mean is **unknown**, Ross proves $E[S^2] = \sigma^2$ for $S^2 = \sum_i (X_i - \bar{X})^2/(n-1)$.

We know,

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2.$$

Taking expectations gives

$$\begin{aligned} (n-1)E[S^2] &= nE[X_1^2] - nE[\bar{X}^2] \\ &= n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right) \\ &= (n-1)\sigma^2. \end{aligned}$$

Therefore $\boxed{E[S^2] = \sigma^2}$, so S^2 is unbiased.

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Key Concept

Known versus unknown mean: If μ is known, $V = \sum_i (X_i - \mu)^2$ satisfies $E[V] = n\sigma^2$, and V/n is unbiased. When μ is unknown and replaced by $\bar{X}$, $E[\sum_i (X_i - \bar{X})^2] = (n-1)\sigma^2$, requiring division by $n - 1$.

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